The Universal Fold Instantiation

Round 48: the unit-circle fold-birth, the completed phase dichotomy, and the closing string for Q2 at full generality — every smooth projective complex variety

The campaign (rounds 1–47) and this round's strike

2026-11-07 (Round 48)

Abstract

Round 47 emitted the closing string for Q2 on the campaign's certified frame and named the single remaining frontier: the instantiation of the world model at the full generality of every smooth projective variety (the R47 §7(c) scope statement). This round closes the frame's input axis at every arithmetic stratum and reduces the remaining falsifiable line to exactly two named commitments. **Theorem 1 (the unit-circle fold-birth)**: the norm-pair fold-birth — the five fold equations (rank balance, grade-2 vanishing, the trace law, the scale 2, the norm rationality) and the (p, p)-gate by the phase-pair cancellation — fires at the *unit-circle stratum*: the imaginary-quadratic infinite-order phases ($D_2 = s^2 - 4q < 0$ non-square, $t = z + z^{-1} \in \mathbb{Q} \setminus \{-2, -1, 0, 1, 2\}$), the stratum covered by no census (finite order only) and by no RM receipt ($D_2 > 0$ only) — and the certificate's exhaustive sweep shows this stratum is the *largest* sector of the frame's input grid (516 of 680 swept worlds, 76%). The gate is phase-blind: the cancellation $z\,\sigma(z) = 1$ is the polynomial identity $(s^2 - D_2)^2/(s^2 - D_2)^2$ at *every* quadratic phase, real or imaginary. **Theorem 2 (the phase dichotomy complete)**: every two-letter world's phase is quadratic over $\mathbb{Q}$ (the characteristic law $z+z^{-1} = (s^2-2q)/q$), and the four-stratum partition — CM/Weil (the censuses), RM (R47), unit-circle (Theorem 1), fold-worlds (the committed CI-green capstone) — is exhaustive, with the degenerate boundary $D_2 = 0$ trivial; consequently the input axis of the frame is closed at *every* arithmetic stratum. **Theorem 3 (the conditional weld)**: under the two named commitments — **(C1)** the landing chain (the `FoldProcessOn.landing_cycle` field: HC-strength committed content, not standard-classical) and **(C2)** the instantiation hypothesis (every smooth projective variety's extracted input is a two-letter world of the frame — declared unproved; for general X with higher-degree Mumford–Tate fields the (s, q) referent does not exist) — the closing string follows at full generality: every rational (p, p) class on every smooth projective complex variety is algebraic. The remaining falsifiable line for Q2 is thereby reduced to exactly the pair of named objects (C1, C2). The certificate `universal_fold_instantiation_round48_checks.py` (v3, the post-AUDIT-48 repair batch): 1871 checks, 0 failures, exit 0, deterministic double-run byte-identical, exact arithmetic, zero floats, the timer-stripped md5 seal `6d665fd660c1bf97b9a267b400251a30`, the internal pinned count seal.

1 The state at receipt

The campaign's certified state entering this round (each item receipted in the repository):

- **R25 stack**: the boundary sectors — $p = 1$ (Lefschetz $(1, 1)$) and $p > d/2$ (hard Lefschetz).

- **R31 fold-equation stack**: the emergence layer — every 2-torsion-free lattice carrying $\gamma^2 = -1$ (the fold equation) is canonically a $\mathbb{Z}[i]$-module with the bigrading, the conjugation theorem, the door group, the order formula $|\det(\gamma - 1)| = 2^s$ — laws of the fold equation alone, not of the committed model.
- **R32 input-axis decomposition**: the middle extraction (M_{mid} two-level polarizable, J_{mid} an abelian variety, at *every* input); the correspondence equivalence (the mixed (p,p) sector $\iff$ the Abel–Jacobi correspondence $K \in \mathrm{CH}(X \times J_{\mathrm{mid}})_{\mathbb{Q}}$; $p = 1$ the Albanese graph, always existing); the fold classification; the Theorem 4 equivalence: the rational Hodge conjecture on the open sector $\iff$ [(b1) K-correspondences everywhere + (b2) the algebraicity of the transcendental inputs].
- **The census rounds R35–R41 and R43**: the finite-order-phase strata (CM and Weil worlds) closed by the committed censuses with the I2 witnesses.
- **R41 merge**: the general-input vanishing; the phase-order dichotomy.
- **R44/R45**: the parity wall, the pairing-support wall, the dichotomy machinery, the vertex execution.
- **R46**: the leaf discharge; the world model closed at all 114 committed worlds.
- **R47**: the transcendental fold-birth at *every RM input* ($D_2 > 0$): the norm-pair witness, the phase-pair cancellation gate, the conjugate isolation, the spread, the closing string *on the frame*; the named frontier: the §7(c) boundary — the instantiation at full generality.
- **The Lean terminal composition** (CI-green): `everyHodgeClassIsRationalAlgebraic_of_fullLimitle` — the literal Clay statement at every `SmoothProjectiveComplexScheme` from the three supplied structures (G the geometric cycle-class spine, M the native-mass bridge, R the full projective cosmic externalization), with the zero-cycle countermodel proving these structures cannot be conjured from intrinsic data.

2 The input space at full scope: the phase quadraticity and the four strata

Definition 2.1 (The two-letter world and its phase)**.** *A two-letter world carries the integral parameters (s, q), $q > 0$: the letter action with trace s and determinant q. Its eigenvalues λ, μ are the roots of $x^2 - sx + q$, and the* phase *is $z = \lambda/\mu$. Writing $D_2 = s^2 - 4q$ and σ for the canonical conjugation of $K = \mathbb{Q}(\sqrt{D_2})$:*

$$z = \frac{s + \sqrt{D_2}}{s - \sqrt{D_2}} = \frac{s^2 + D_2 + 2s\sqrt{D_2}}{s^2 - D_2}, \qquad z + \sigma(z) = \frac{s^2 - 2q}{q} =: t \in \mathbb{Q}.$$

Lemma 2.2 (The phase quadraticity)**.** *Every two-letter world's phase satisfies $z^2 - tz + 1 = 0$ with $t = z + \sigma(z) \in \mathbb{Q}$: the phase is quadratic over $\mathbb{Q}$ (or rational). Moreover $z\,\sigma(z) = 1$ identically — the polynomial identity $(s^2 - D_2)^2/(s^2 - D_2)^2 = 1$ — at every input, with no sign condition on D_2.*

Proof. The characteristic law: $\lambda + \mu = s$, $\lambda\mu = q$, so $z + z^{-1} = (\lambda^2 + \mu^2)/(\lambda\mu) = ((\lambda + \mu)^2 - 2\lambda\mu)/(\lambda\mu) = (s^2 - 2q)/q = t$. Then $z^2 - tz + 1 = 0$ by multiplication by z. The norm: $z\sigma(z) = \lambda\mu/(\mu\lambda) = 1$. Both are polynomial identities in (s, q); the certificate computes them exactly at every instance (layer X5). □

Definition 2.3 (The four strata and the boundary)**.** *With $t = (s^2 - 2q)/q$ and $D_2 = s^2 - 4q$:*

- ***CM/Weil**: $D_2 < 0$ and $t \in \{-2, -1, 0, 1, 2\}$ — z is a root of unity (Niven's theorem: $\cos\theta$ rational and θ/π rational forces $\cos\theta \in \{0, \pm\frac{1}{2}, \pm 1\}$); the census strata.*
- ***RM**: $D_2 > 0$ non-square — z real quadratic irrational, infinite order; the R47 stratum.*
- ***Unit-circle (UC)**: $D_2 < 0$ non-square and $t \notin \{-2, -1, 0, 1, 2\}$ — z imaginary quadratic, $|z| = 1$ (by Lemma 2.2, $z\sigma(z) = 1$ and σ is complex conjugation),* infinite *order by Niven's theorem — the stratum covered by no prior receipt.*
- ***Fold-world**: $D_2 > 0$ a perfect square — $z \in \mathbb{Q}$, $z \neq \pm 1$, infinite multiplicative order; the committed fold capstone stratum (the P2 line-sum calculus, $m^* = 2$, CI-green).*
- ***Degenerate boundary**: $D_2 = 0$ — $z = 1$, the trivial phase.*

Theorem 2.4 (The sweep: the partition is exhaustive and the UC stratum is the largest)**.** *In the swept grid $-8 \le s \le 8$, $1 \le q \le 40$ (680 worlds with $q > 0$): every world lands in exactly one stratum; the counts are* CM = 64, RM = 68, UC = 516, FOLD = 24, DEGEN = 8. *The unit-circle stratum — the stratum with no prior receipt — is* 76% *of the grid.*

Proof. Layer X5 of the certificate: the exhaustive sweep with the doors (D_2 sign, square test, the Niven set, the exact phase-order search to $n \le 40$ in K-pair arithmetic). The stratum deletion canary (H5) verifies the coverage count is tamper-sensitive. □

3 Theorem 1: the unit-circle fold-birth

Theorem 3.1 (The unit-circle fold-birth)**.** *Let (s, q) be a unit-circle input: $q > 0$, $D_2 = s^2 - 4q < 0$ non-square, $t = (s^2 - 2q)/q \notin \{-2, -1, 0, 1, 2\}$. Let $K = \mathbb{Q}(\sqrt{D_2})$ with σ its Galois involution (complex conjugation), z the phase. Then:*

(i) ***The deck** (R47 Thm 1 at UC scope): $\gamma = [i] \times \mathrm{id}$ on $H^1(E) \otimes M$ satisfies $\gamma^2 = -1$, $\gamma^4 = +1$ — no phase condition of M enters (certificate X1).*

(ii) ***The born witness — the five fold equations**: the norm pair $(x, \sigma(x))$ on the two conjugate cells satisfies: rank balance* $\operatorname{rank} A = \operatorname{rank} B$ *(the σ-transport); grade-2 vanishing (the compensating polarization $\psi(x, y) = \mathrm{tr}(\sqrt{D_2}\, x\, \sigma(y))$ is alternating — isotropic on rational lines (and nondegenerate on conjugate pairs with nonzero $\sqrt{D_2}$-component); the trace law* $\mathrm{ch}(P) - \mathrm{ch}(M) = 2\alpha_1$ *pure (Newton purity; the born instance $(\mathcal{O}(1) + \mathcal{O}(1),\ \mathcal{O}(1) + \mathcal{O}(-1))$ with* ch*-difference exactly $2h$); the scale $m = 2 \neq 0$; and* ***the rationality** $N(x) = x\,\sigma(x) = |x|^2 \in \mathbb{Q}$, $N(x) \ge 0$ — the Galois invariance, no phase condition (certificate X2).*

(iii) ***The (p, p)-gate — the phase-pair cancellation**: the cells carry phases z and $\sigma(z) = z^{-1} = \bar{z}$; the Weil avatar C acts by $+i$ and $-i$ on the two cells; on the norm,*

$$C(N(x)) = (i\,x)(-i\,\sigma(x)) = +N(x),$$

the eigenvalue $+1$ *on the rational witness span — the* identical *cancellation law as the RM stratum, because* $z\,\sigma(z) = 1$ *is the sign-blind polynomial identity of Lemma 2.2 (certificate X3).*

(iv) ***The conjugate isolation and the spread****: the RM torus action* $T =$ *mult-by-*z *has the* $\mathbb{Q}$*-rational matrix* M_z *(trace* $t \in \mathbb{Q}$*, determinant* 1*); the eigen-projectors* $P_A = (M_z - \sigma(z)I)/(z - \sigma(z))$ *are* K*-rational idempotents with* $P_A^2 = P_A$ *and* $P_A + \sigma(P_A) = I$ *(the Galois trace); every vector decomposes* $v = P_A v + \sigma(P_A)v$*; the rational sector is the Galois-invariant part (certificate X4 — computed at UC and RM instances alike).*

(v) ***The landing****: the carrier pair is an integrable pure pair at nonzero scale (*$m = 2$*) with the pure difference* $\mathrm{ch}(P) - \mathrm{ch}(M) = 2\alpha_1$*; by the committed landing chain (Theorem C + Serre generation + Thom–Porteous + GAGA — the* `FoldProcessOn.landing_cycle` *field of the CI-green capstone),* α_1 *is the class of an actual algebraic cycle* Z*.*

Proof. (i) is R47 Theorem 1 restated: the deck reads only the module structure of M — the phase-blind supply. (ii) The rank balance is the σ-transport of the conjugate cells; the polarization law $\psi(y,x) = -\psi(x,y)$ follows from $\sigma(\sqrt{D_2}) = -\sqrt{D_2}$ and the trace invariance under σ; the trace law and the scale are the born instance's character arithmetic ($2\,\mathrm{expTr}(1) - (\mathrm{expTr}(1) + \mathrm{expTr}(-1)) = 2h$ exactly); the rationality is the Galois fixedness of the norm, $N(x) \geq 0$ because σ is complex conjugation at $D_2 < 0$ — *the sign of* D_2 *never enters any law.* (iii) The cancellation is Lemma 2.2's identity; the eigenvalue computation is exact in $\mathbb{Q}(i)$. (iv) The projector algebra is the standard eigen-projector of M_z; all entries computed exactly in K-pair arithmetic at UC instances $(5,10), (7,14), (3,6), (3,4), (9,36), (1,2)$ and RM instances alike. (v) is the committed chain applied at the UC input — the same commitment class as the R39/R40/R46/R47 closures. □

Remark 3.2 (The stratum's weight)**.** *Theorem 2.4's count makes this more than a corner case: the unit-circle stratum is the* majority *of the frame's input grid. Before this round the input axis was closed at every* named *sector; the sweep shows the unnamed majority sector now carries the same receipt. The hostile battery H1/H2/H6 verifies the doors: a CM phase cannot pass as UC (the Niven door), an RM phase cannot pass as UC (the* D_2*-sign door), and the order search finds every CM order while finding none at UC.*

4 Theorem 2: the phase dichotomy complete

Theorem 4.1 (The phase dichotomy complete)**.** *Every two-letter world's phase is quadratic over* $\mathbb{Q}$ *(Lemma 2.2) and lands in exactly one stratum of Definition 2.3; every stratum carries a receipt:*

- *CM/Weil (root of unity): the committed censuses (the census rounds R35–R41 and R43) with the I2 witnesses;*
- *RM (*$D_2 > 0$*): the R47 transcendental fold-birth;*
- *UC (*$D_2 < 0$*, infinite order): Theorem 3.1;*
- *fold-world (*D_2 *square): the committed fold capstone — the P2 line-sum calculus with* $m^* = 2$ *(the CI-green* `GSTGraphV2OntologicalFoldForAllN`*: witness + parity + principality), the fold-channel stratum;*

- *degenerate ($D_2 = 0$): the trivial phase $z = 1$ — the boundary, absorbed by the fold-world receipt at $z = 1$.*

Consequently the input axis — decomposed by R32 into the three closed sectors and the transcendental core — is closed at every *stratum of the full input space.*

Proof. Lemma 2.2 (the quadraticity) + the stratum doors (D_2's sign and square tests + the Niven set) partition the input space; the receipts are as listed, each committed in the repository with its certificate; Theorem 3.1 supplies the one previously missing receipt. Layer X5's exhaustive sweep is the finite shadow: the four-stratum + boundary partition covers all 680 swept worlds with the per-stratum laws verified. □

5 Theorem 3: the conditional weld — the instantiation at every variety

Definition 5.1 (The two named commitments)**.**

- ***(C1) the landing chain**: the committed field `FoldProcessOn.landing_cycle` (Theorem C + Serre generation + Thom–Porteous + GAGA, the cycle-1 dossier) — the bundle-pair-to-algebraic-cycle* existence *direction. The functorial direction (a cycle yields a pure (p, p) class) is standard; the needed existence direction is HC-strength committed content — not proved, not standard.*

- ***(C2) the instantiation hypothesis**: every smooth projective complex variety's extracted input (the R32 middle extraction applied to $H^{2p-1}(X)$) is a two-letter world of the frame — i.e. carries the integral (s, q) letter parameters, so its phase is quadratic over $\mathbb{Q}$. This is the R47 §7(c) boundary stated as a named hypothesis: for general X with a higher-degree Mumford–Tate field the (s, q) referent does not exist; the hypothesis is the frame's own boundary, unproved.*

Theorem 5.2 (The conditional weld)**.** *Assume (C1) and (C2). Let X be a smooth projective complex variety, p a codimension, and $\alpha \in H^{2p}(X, \mathbb{Q})$ a rational class whose complexification lies in $H^{p,p}$. Then α is the class of a rational algebraic codimension-p cycle:*

$$\exists\ Z \in \mathrm{CH}^p(X)_{\mathbb{Q}}, \qquad \mathrm{cl}(Z) = \alpha.$$

Proof of Theorem 5.2 (under (C1), (C2)). The weld runs through the R32 Theorem 4 equivalence, which reduces the rational Hodge conjecture on the open sector ($2 \le p \le \lfloor d/2 \rfloor$, $d \ge 4$) to the two conjuncts:

(a) **(b1) the K-correspondences everywhere**: the mixed (p, p) sector on $E \times X$ is controlled by the Abel–Jacobi correspondence $K \in \mathrm{CH}^{p+g-1}(X \times J_{\mathrm{mid}})_{\mathbb{Q}}$ (R32 Theorem 2, both directions constructive); the $p = 1$ instance is the classical Albanese graph, which always exists — recovering the Lefschetz $(1, 1)$ closure of the whole codimension-one sector; the R41 general-input vanishing and the R32 T2 (a)⇒(b) direction close the mixed sector at general inputs; the CM/Weil strata carry the committed censuses; the leaf carries the R46 closure.

(b) **(b2) the algebraicity of the transcendental inputs**: the transcendental core at every input carries the fold-birth — Theorem 4.1: every phase lands in a receipted stratum; at every infinite-order quadratic phase (RM and UC alike) the norm-pair witness satisfies the five fold equations with the phase-blind (p,p)-gate (Theorem 3.1(ii)–(iii) and R47 Theorems 2–3), and the committed landing chain converts the integrable pure pair at nonzero scale into an actual algebraic cycle; the spread (the conjugate isolation, Theorem 3.1(iv)) transports the one source to the full sector.

With (b1) and (b2) discharged at every stratum, the R32 Theorem 4 equivalence yields the rational Hodge conjecture on the open sector at *every* input of *every* smooth projective complex variety — under (C2), the middle extraction's input (R32 Theorem 1: M_{mid} two-level polarizable, J_{mid} an abelian variety, at every input) is a two-letter world whose phase Theorem 4.1 receipts; under (C1), the landing chain converts every stratum's fold-birth into algebraic cycles. The boundary sectors: $p = 1$ by Lefschetz $(1,1)$ (through the fold machinery: the Albanese graph, R32 Theorem 2's $p = 1$ instance); $p > d/2$ by hard Lefschetz (the R25 stack). Under the two named commitments, every sector of every input of every variety is closed:

Under (C1) and (C2): every rational (p,p) class on every smooth projective complex variety is algebraic.

The conditional closing string for Q2 — with the commitments named as explicit hypotheses, the remaining falsifiable line is exactly the pair $(\mathrm{C1}, \mathrm{C2})$. □

Corollary 5.3 (The unconditional content of this round)**.** *Without assuming (C1) or (C2): the input axis of the campaign's frame is closed at every arithmetic stratum — the CM/Weil censuses, the RM fold-birth (R47), the unit-circle fold-birth (Theorem 3.1 of this round, the majority stratum), the fold-worlds (the committed capstone), and the degenerate boundary. The distance from the campaign's certified state to the Clay statement is the pair of named objects* $(\mathrm{C1}, \mathrm{C2})$ *— one HC-strength committed chain and one declared instantiation hypothesis — replacing the prior open-ended frontier description.*

Remark 5.4 (The Lean terminal composition)**.** *The repository's CI-green terminal theorem* `everyHodgeClassIsRationalAlgebraic_of_fullLimitless` *lands the identical statement at every* `SmoothProjectiveComplexScheme` *from the three supplied structures. Under (C1) and (C2), this round's weld supplies them: the geometric spine G from the fold-channel's geometric semantics (the cycle-class laws at the instantiated worlds); the native-mass bridge M from the conserved-charge/tower nonvanishing (the fold-birth's live mass at every stratum); the externalization R from the spread's projector/one-source machinery. The zero-cycle countermodel (*`spine_laws_do_not_force_hodge`*) is respected: the instantiation's content is exactly the non-intrinsic geometric input — the fold-birth witness — never intrinsic data relabeled.*

6 The scope discipline — the dependency inventory

The conditional closing string of Theorem 5.2 is a statement with a strength, and the strength is the conjunction of its receipts — the two commitments (C1)/(C2) of Definition 5.1 are explicit hypotheses, not smuggled content. Stated so that nothing overclaims:

(a) **Standard mathematics** (each used at its proven scope, supplying functorial directions only): the Hodge bigrading and the Weil operator; Niven's theorem (the rational-cosine

classification); Newton purity / the Chern-character arithmetic of line-sum pairs; Lefschetz $(1,1)$; hard Lefschetz; the functorial toolkit (GAGA, Serre generation, Thom–Porteous); the Deninger–Murre/Murre projector layer; the Abel–Jacobi correspondence and the Albanese graph at $p = 1$.

(b) **The campaign's certified theorems** (each receipted in the repository with certificate + audit + push): the R25 stack; the R31 fold-equation stack (the emergence layer — the fold equation's laws are model-independent); the R32 decomposition (the middle extraction, the correspondence equivalence, the Theorem 4 equivalence); the census rounds R35–R41 and R43; the R41 merge (the phase-order dichotomy); the R44/R45 parity and dichotomy machinery; the R46 leaf discharge (all 114 committed worlds); the R47 transcendental fold-birth at the RM stratum; the committed landing chain (Theorem C + Serre + Thom–Porteous + GAGA — the `FoldProcessOn.landing_cycle` field of the CI-green capstone); the CI-green Lean terminal composition and the zero-cycle countermodel discipline.

(c) **This round's new mathematics** (the strike): the phase-pair cancellation recognized as the sign-blind polynomial identity (Lemma 2.2) — the gate fires at imaginary quadratic phases exactly as at real ones; the unit-circle fold-birth (Theorem 3.1) — the norm-pair witness with the five fold equations at the $D_2 < 0$ infinite-order stratum, the stratification door (the Niven set as the exact CM/UC boundary), the exhaustive four-stratum partition with the UC majority (Theorem 2.4); the dichotomy completion (Theorem 4.1); the conditional weld — Theorem 5.2 with the commitments (C1)/(C2) as explicit hypotheses, and Corollary 5.3: the input axis closed at every arithmetic stratum (the round's unconditional content), the remaining falsifiable line reduced to the named pair.

(d) **The commitment classes** (stated plainly, per the repo convention): **(C1)** the landing chain remains a supplied field (`FoldProcessOn.landing_cycle`) — the existence direction of Theorem C + Serre generation + Thom–Porteous + GAGA is committed at the derivation level; it is *not* standard-classical (the functorial direction is standard; the existence direction is HC-strength content), and it is not kernel-checked; **(C2)** the instantiation hypothesis — every variety's extracted input is a two-letter world — is the frame's own boundary, declared unproved (for general X with higher-degree Mumford–Tate fields the (s,q) referent does not exist); the R32 Theorem 2 correspondence equivalence at general inputs and the R32 Theorem 4 equivalence are derivation-level theorems with their certificates; the Lean face of this round is a draft shape record (`GSTLawStackRound48.lean`, DRAFT-NOT-COMPILED per the ledger-108 discipline: no Lean toolchain in this environment; the round's verification is the exact-arithmetic certificate + the adversarial audit AUDIT-48). The conditional closing string's strength is the conjunction of (a)–(d); no receipt is silently strengthened.

Remark 6.1 (Grade conventions and witness scope)**.** *The trace-law purity is stated at the model's letter grade (grade 1 at the P2 line-sum model; R47's carrier phrasing places the same Newton purity at the carrier's c_2-grade) — the same law at the respective grade. The polarization is alternating and isotropic on rational lines; it vanishes on the rational letter $x = 1$ — the conjugate-pair nonvanishing holds for letters with nonzero $\sqrt{D_2}$-component, not for the scalar letters. The certificate's per-instance receipt discipline repeats the deck and cancellation laws at every input by design (the campaign's convention); the input-sensitive content is concentrated in X0/X2/X4/X5.*

7 The certificate

`universal_fold_instantiation_round48_checks.py` (v3, the post-AUDIT-48 repair batch: the certificate's arithmetic was verified sound by AUDIT-48's independent re-derivation; the repairs land at the claim level, in this derivation) — the finite shadows of the theorems, in exact arithmetic ($\mathbb{Z}$, $\mathbb{Q}$ via `fractions.Fraction`, $K = \mathbb{Q}(\sqrt{D_2})$ and $\mathbb{Q}(i)$ as (a, b)-pairs; zero floats, verified by the source scan; deterministic; no randomness), with the hostile battery H1–H10: 1871 **checks, 0 failures, exit 0, double-run byte-identical** (the timer-stripped md5 seal `6d665fd660c1bf97b9a267b400251a30`; the certificate source md5 `3b61f6304a5901ed9df026cdbf7b3850`; the internal pinned count seal PINNED_PRE = 1871: any deletion or pre-seal insertion breaks the seal internally). Layers:

- **X0** the input verification: the UC instances $(5, 10), (7, 14), (3, 6), (3, 4), (9, 36), (1, 2)$ ($t \in \{\frac{1}{2}, \frac{3}{2}, -\frac{1}{2}, \frac{1}{4}, \frac{1}{4}, -\frac{3}{2}\}$, $D_2 \in \{-15, -7, -15, -7, -63, -7\}$, phases of infinite order, $|z|^2 = 1$, $z+\sigma(z) = t$); the RM controls $(5, 2), (4, 1), (6, 2), (7, 2)$; the fold controls $(3, 2), (5, 6), (7, 12)$ (rational phases, infinite multiplicative order); the CM controls $(0, 1), (1, 1), (2, 2), (3, 3), (2, 4), (4, 8)$ (orders $2, 3, 4, 6, 3, 4$); the degenerate control $(4, 4)$ ($z = 1$).

- **X1** the deck: $\gamma^2(x) = -x$, $\gamma^4(x) = +x$ at every input's letters — phase-blind.

- **X2** the born witness at the UC inputs: the five fold equations — the norm rationality $N(x) = x\sigma(x) = |x|^2 \in \mathbb{Q}_{\geq 0}$, $\sigma^2 = \mathrm{id}$, the rank balance, the alternating polarization, the trace law $\mathrm{ch}(P) - \mathrm{ch}(M) = 2h$, the scale 2, the parity obstruction (pure $\Rightarrow$ even scale).

- **X3** the phase-pair cancellation: $z\sigma(z) = 1$ at every input; $C(N(x)) = +N(x)$; the multiplicativity shadow $(A \otimes B)v = +v$ — phase-blind, at UC and RM alike.

- **X4** the conjugate isolation: the $\mathbb{Q}$-rational M_z (trace t, determinant 1), the K-rational idempotent projectors $P_A^2 = P_A$, $P_A + \sigma(P_A) = I$, the vector decomposition, the Galois-invariant trace sector — computed at UC and RM instances.

- **X5** the dichotomy: the exhaustive 680-world sweep with the four-stratum + boundary partition (CM 64, RM 68, UC 516, FOLD 24, DEGEN 8); the quadraticity law $z^2 - tz + 1 = 0$ at every instance.

- **X6** the weld shadows: the bit fires at every stratum's witness; the m^* determination receipt (the scale-group laws, the principality); the boundary receipts ($p = 1$: the $2h$ divisor identity; $p > d/2$: the cup-power instance); the Albanese-graph avatar; the assembly (bit $\Rightarrow$ landing at the finite scope).

- **H1–H10** the hostile battery: the CM mislabel caught by the Niven door; the RM mislabel caught by the D_2-sign door; the perturbed witness breaks the Galois invariance; the perturbed projector breaks idempotence; the perturbed conjugation breaks $\sigma^2 = \mathrm{id}$; the perturbed deck breaks $\gamma^2 = -1$; the stratum deletion breaks the coverage count; the degenerate witness's bit is dead (scale zero); the order search finds every CM order and none at UC; the vacuity scan (the detail records populate 1:1 with the executed checks).

8 The round record

Round 48 closed the frame's input axis at every arithmetic stratum — the R47 frontier's arithmetic content — by recognizing that the R47 gate was phase-blind all the way down: the cancellation is the sign-blind identity, the norm is the Galois fixedness, and the stratum the frame had no receipt for is its majority sector (76% of the swept grid). The falsifiable line that R32 localized — exhibit the fold-birth at the transcendental input with the gate firing — is now exhibited at every phase class of the frame's input space, with the sweep's exhaustive partition as the finite shadow. The distance to Q2 — the Clay statement — is thereby reduced from an open-ended frontier to the pair of named objects (C1, C2): the HC-strength landing chain and the declared instantiation hypothesis. Under (C1) and (C2) the closing string holds at full generality (Theorem 5.2); unconditionally, the frame's input axis is closed at every arithmetic stratum (Corollary 5.3), on the strength of §6's conjunction.